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Adjusted location-invariant U-tests for the covariance matrix with elliptically high-dimensional data

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Abstract

This paper analyzes several covariance matrix U-tests, which are constructed by modifying the classical John-Nagao and Ledoit-Wolf tests, under the elliptically distributed data structure. We study the limiting distributions of these location-invariant test statistics as the data dimension p may go to infinity in an arbitrary way as the sample size n does. We find that they tend to have unsatisfactory size performances for general elliptical population. This is mainly because such population often possesses high-order correlations among their coordinates. Taking such kind of dependency into consideration, we propose necessary corrections for these tests to cope with elliptically high-dimensional data. For computational efficiency, alternative forms of the new test statistics are also provided. We derive the universal (n, p) asymptotic null distributions of the proposed test statistics under elliptical distributions and beyond. The powers of the proposed tests are further investigated. The accuracy of the tests is demonstrated by simulations and an empirical study.

KEYWORDS

covariance testing, elliptical distributions, large p small n , U-statistics

1 | INTRODUCTION

Inference for covariance matrices plays a fundamental role in multivariate statistics. Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be independent and identically distributed p -dimensional random vectors, whose covariance matrix is $\Sigma = \text{var}(\mathbf{x}_i)$. In this article, we consider two common testing problems on the covariance matrix Σ : one is identity test and the other is sphericity test. The hypothesis for identity test is

$$H_0 : \Sigma = \mathbf{I}_p \quad \text{versus} \quad H_1 : \Sigma \neq \mathbf{I}_p, \quad (1)$$

where $\mathbf{I}_p$ is the p -dimensional identity matrix. For a given invertible matrix Σ_0 , testing the null hypothesis that Σ is equal to Σ_0 can be converted to testing (1) through multiplying the data by $\Sigma_0^{-1/2}$. The hypothesis for sphericity test is

$$\tilde{H}_0 : \Sigma = \sigma^2 \mathbf{I}_p \quad \text{versus} \quad \tilde{H}_1 : \Sigma \neq \sigma^2 \mathbf{I}_p, \quad (2)$$

where σ^2 is an unknown but finite positive constant. This test is to examine whether the covariance matrix Σ is proportional to the identity matrix $\mathbf{I}_p$. The identity and sphericity testing can be applied to biology, geology, economics and finance, among others. See Mardia et al. (2000), Hallin and Paindaveine (2006), Chen et al. (2010) and Yang et al. (2021) for example. In this article, we focus on the two testing problems under the universal (n, p) -asymptotic framework (Paindaveine & Verdebout, 2016) where the dimension p diverges to infinity freely as $n \rightarrow \infty$.

To test (1) and (2) under the fixed dimension $p < n$, the likelihood ratio tests (Anderson, 2003) are perhaps the most commonly used methods. Sugiura and Nagao (1968) modified the biased likelihood ratio criteria and proposed the tests based on

$$\begin{aligned} \lambda_1^* &= \exp\{p(n-1)/2\} \{\det(\mathbf{S})\}^{(n-1)/2} \exp\{-(n-1)\text{tr}(\mathbf{S})/2\}, \\ \lambda_2^* &= \{\det(\mathbf{S})\}^{(n-1)/2} \{p^{-1}\text{tr}(\mathbf{S})\}^{-p(n-1)/2}, \end{aligned}$$

for (1) and (2) under the multivariate normal assumption, where $\exp(\cdot)$ is the exponential function, $\det(\cdot)$ is the determinant of a matrix, $\text{tr}(\cdot)$ is the trace of a matrix, $\mathbf{S} = (n-1)^{-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$ is the sample covariance and $\bar{\mathbf{x}} = n^{-1} \sum_{i=1}^n \mathbf{x}_i$ is the sample mean. John (1971, 1972) and Nagao (1973) focused on the limiting variances of $-2 \log(\lambda_1^*)$ and $-2 \log(\lambda_2^*)$, and considered the test statistics

$$V_{n,p} = \text{tr}\{(\mathbf{S} - \mathbf{I}_p)^2\}/p, \quad (3)$$

$$U_{n,p} = \text{tr}([\mathbf{S}/\{(1/p)\text{tr}(\mathbf{S})\} - \mathbf{I}_p]^2)/p. \quad (4)$$

The locally most powerful property of the tests has been derived with multivariate normal distributions. To make the tests valid for the elliptical distributed data, Muirhead and Waternaux (1980) suggested incorporating the kurtosis into $-2 \log(\lambda_2^*)$ and $U_{n,p}$, Hallin and Paindaveine (2006) and Sirkiä et al. (2009) constructed the nonparametric tests with multivariate sign- or rank-based covariance matrices.

A prominent feature of collected data nowadays is that p can be comparable with, or even much larger than n . In this situation, the sample covariance matrix $\mathbf{S}$ is no longer consistent (Bai & Silverstein, 2005). Therefore, unlike the conventional formulations of John (1971, 1972), Nagao (1973) and Anderson (2003), tests for the high-dimensional hypotheses (1) and (2) would not be directly constructed based on the sample covariance matrix. There have been advances in constructing consistent covariance estimators for high-dimensional data via the regularization

(Huang et al., 2006; Rothman et al., 2010), truncation (Bickel & Levina, 2008a; Cai et al., 2010) or thresholding (Bickel & Levina, 2008b; Rothman et al., 2009) methods. Despite these developments, it is still challenging to transform these results to test procedures on high-dimensional covariance matrices.

To accommodate large p , Bai et al. (2009), Wang and Yao (2013), and Yang et al. (2021) corrected the likelihood ratio tests when $p/n \rightarrow c \in (0, 1)$. Ledoit and Wolf (2002) rechecked the asymptotic properties of $U_{n,p}$ and $V_{n,p}$ for normally distributed data. They proved that with $p/n \rightarrow c$ for a finite constant $c > 0$, the sphericity test based on $U_{n,p}$ can be directly applied, whereas the identity test built on $V_{n,p}$ needs some modifications. Birke and Dette (2005) derived the asymptotic null distribution of the Ledoit-Wolf test statistic when both $n, p \rightarrow \infty$ and $p/n \rightarrow \infty$. Along this research line, many testing procedures have been developed, see Srivastava (2005, $n = O(p^\delta)$, $0 < \delta \leq 1$), Fisher et al. (2010, $p/n \rightarrow c > 0$), Fisher (2012, $p/n \rightarrow c \geq 1$), Zhang et al. (2013, $p = o(n^{\{\delta + \min(2, \delta)\}/\{4(2+\delta)\}})$, $\delta > 0$), Srivastava et al. (2014, $n = O(p^\delta)$, $1/2 < \delta < 1$), Tian et al. (2015, $p/n \rightarrow c > 0$), Hu et al. (2019, $p/n \rightarrow c > 0$), Yang et al. (2021, $p/n \rightarrow c > 0$) and so on. However, the aforementioned tests all impose strong assumptions on the relationships between n and p to establish their (n, p) -asymptotic results.

In the literature, there only exist a few works that allow for universal (n, p) -asymptotics (Paindaveine & Verdebout, 2016) to test (1) and (2). Chen et al. (2010) and Ahmad (2016) provided location-invariant U-tests without specifying the growth rate of p with respect to n . The asymptotic normality of their test statistics is built under the pseudo-independence component models:

- There exists a p -dimensional random vector $\mathbf{z}_i = (Z_{i1}, \dots, Z_{ip})^T$ such that

$$\mathbf{x}_i = \mathbf{\Gamma} \mathbf{z}_i + \mathbf{u}, \text{ for } i = 1, \dots, n, \quad (5)$$

where $\{\mathbf{z}_i\}_{i=1}^n$ are p -dimensional independent and identically distributed random vectors that satisfy (i) $E(\mathbf{z}_i) = \mathbf{z}_p$, where $\mathbf{z}_p$ denotes the p -dimensional zero vector; (ii) $\text{var}(\mathbf{z}_i) = \mathbf{I}_p$; (iii) the 8th moment of Z_{ik} is uniformly bounded, and $E(Z_{ik}^4) = 3 + \Delta$ for some finite constant Δ for $k = 1, \dots, p$; (iv) for any positive integers q , and $\ell_v \geq 0$ with $\sum_{v=1}^q \ell_v \leq 8$,

$$E(Z_{ik_1}^{\ell_1} Z_{ik_2}^{\ell_2} \dots Z_{ik_q}^{\ell_q}) = E(Z_{ik_1}^{\ell_1}) E(Z_{ik_2}^{\ell_2}) \dots E(Z_{ik_q}^{\ell_q}), \quad (6)$$

where $k_1, \dots, k_q$ are distinct indices. Besides, $\mathbf{u} = E(\mathbf{x}_i)$ is a p -dimensional constant vector, $\mathbf{\Gamma}$ is a $p \times p$ matrix such that $\mathbf{\Gamma} \mathbf{\Gamma}^T = \mathbf{\Sigma}$.

The data structure (5) encompasses the useful distributions considered by Bai and Saranadasa (1996), Chen and Qin (2010), and Ilmonen and Paindaveine (2011). However, the dependence assumption in (6) still excludes some important distributions. In particular, we are interested in the distributions from elliptical family (Fang et al., 1990), including multivariate t -distribution, multivariate logistic distribution, symmetric multivariate Laplace distribution, and so on. This elliptical family of distributions admits the following representation:

- A random vector $\mathbf{x}_i$ follows an elliptical distribution if and only if it has a stochastic representation (Fang et al., 1990):

$$\mathbf{x}_i = \Lambda R_i \mathbf{u}_i + \mathbf{u}, \quad (7)$$

where Λ is a $p \times p$ matrix such that $p^{-1} E(R_i^2) \Lambda \Lambda^T = \mathbf{\Sigma}$, $R_i > 0$ is a scalar variable that denotes the radius of $\mathbf{x}_i$, and $\mathbf{u}_i$ is a random direction uniformly distributed on the unit sphere in $\mathbb{R}^p$, which is independent of R_i .

Let $\tilde{Z}_{ik} = (R_i \mathbf{u}_i)_k$ for $k = 1, \dots, p$. In general, even if the population correlation matrix is an identity matrix, it can be verified that

$$E(\tilde{Z}_{ik_1}^{\ell_1} \tilde{Z}_{ik_2}^{\ell_2} \cdots \tilde{Z}_{ik_q}^{\ell_q}) \neq E(\tilde{Z}_{ik_1}^{\ell_1}) E(\tilde{Z}_{ik_2}^{\ell_2}) \cdots E(\tilde{Z}_{ik_q}^{\ell_q}), \quad (8)$$

for some scalar variable $R_i > 0$, positive integers q , and $\ell_v \geq 0$ with $\sum_{v=1}^q \ell_v \leq 8$. As explained by Equations (6) and (8), the dependence relations are very different in the two settings. In this sense, a random vector from the family given by (7) has strongly dependent coordinates.

The elliptically distributed data are frequently seen in various scientific areas, for example, biology, economics, and finance. The tests that cannot work on elliptical distributions would have a limited scope of application in practice. To overcome this limitation, Paindaveine and Verdebout (2016) proposed a universal sign test for the sphericity of elliptical populations with a known location parameter. However, the center $\mathbf{u}$ is usually unknown in practice and needs to be estimated. Zou et al. (2014) found that if the location parameter is estimated by the rotation-equivariant spatial median (Möttönen & Oja, 1995), the test of Paindaveine and Verdebout (2016) would have a nonnegligible bias term in high dimension. To guarantee the accurate size performances, Zou et al. (2014) introduced a bias-corrected sign-based test, which allows p to increase to infinity at most as fast as n^2 .

In this article, we develop adjusted location-invariant U-tests for the hypotheses (1) and (2) under the elliptical distributions. For this purpose, we first examine how the U-tests of Chen et al. (2010) and Ahmad (2016) perform under the elliptical setting (7). It is shown that they are not valid for general elliptical population, producing tests with type I error rates that are far away from the nominal levels. This is attributed to the fact that such population often exhibits high-order correlations among their coordinates, as indicated in (8). Toward the theoretical analysis of the tests, we have addressed the difficult problem raised in Zou et al. (2014, p. 233). Taking such kind of dependency into consideration, we develop necessary corrections for these location-invariant tests to make them robust with respect to elliptically high-dimensional data. The universal (n, p) -asymptotic null distributions of our identity and sphericity test statistics are established. The powers of the proposed tests for the high-dimensional scenarios of alternatives are also evaluated.

The rest of the paper is organized as follows. In Section 2, we revisit the location-invariant U-tests for the data structure (5) and study their performances under elliptical distributions. In Section 3, we introduce the proposed test statistics when the underlying distribution is elliptical. We further establish the asymptotic properties under two different kinds of nonparametric distribution in Section 4. In Section 5, we present the results of simulations and an empirical study to check the validity of our asymptotic results. Section 6 concludes the article. All the technical details are deferred to the Appendix S1.

2 | THE LOCATION-INVARIANT U-TESTS AND THEIR CONVERGENCES UNDER ELLIPTICAL DISTRIBUTIONS

2.1 | Several location-invariant U-statistics

Based on the work of Ledoit and Wolf (2002), Chen et al. (2010), and Ahmad (2016) advocated using the cross-validation technique to construct the identity and sphericity test statistics.

Denote $(n)_m = n(n-1)\dots(n-m+1)$ for $1 \leq m \leq n$. Let

$$\begin{aligned} Y_{1,n,p} &= (n)_1^{-1} \sum_{i_1=1}^n \mathbf{x}_{i_1}^T \mathbf{x}_{i_1}, & Y_{2,n,p} &= (n)_2^{-1} \sum_{(i_1, i_2)}^n (\mathbf{x}_{i_1}^T \mathbf{x}_{i_2})^2, \\ Y_{3,n,p} &= (n)_2^{-1} \sum_{(i_1, i_2)}^n \mathbf{x}_{i_1}^T \mathbf{x}_{i_2}, & Y_{4,n,p} &= (n)_3^{-1} \sum_{(i_1, i_2, i_3)}^n \mathbf{x}_{i_1}^T \mathbf{x}_{i_2} \mathbf{x}_{i_2}^T \mathbf{x}_{i_3}, \\ Y_{5,n,p} &= (n)_4^{-1} \sum_{(i_1, i_2, i_3, i_4)}^n \mathbf{x}_{i_1}^T \mathbf{x}_{i_2} \mathbf{x}_{i_3}^T \mathbf{x}_{i_4}. \end{aligned}$$

We use the notations

$$\sum_{(i_1, i_2)}^n, \sum_{(i_1, i_2)}^n, \sum_{(i_1, i_2, i_3)}^n, \sum_{(i_1, i_2, i_3, i_4)}^n \quad \text{and} \quad \sum_{(i_1, i_2, i_3, i_4, i_5)}^n,$$

throughout to denote the summations over mutually different indices. Then, consider the unbiased estimators of $\text{tr}(\Sigma)$ and $\text{tr}(\Sigma^2)$, respectively,

$$\begin{aligned} T_{1,n,p} &= \{2(n)_2\}^{-1} \sum_{(i_1, i_2)}^n (\mathbf{x}_{i_1} - \mathbf{x}_{i_2})^T (\mathbf{x}_{i_1} - \mathbf{x}_{i_2}) = Y_{1,n,p} - Y_{3,n,p} \quad \text{and} \\ T_{2,n,p} &= \{4(n)_4\}^{-1} \sum_{(i_1, i_2, i_3, i_4)}^n \{(\mathbf{x}_{i_1} - \mathbf{x}_{i_2})^T (\mathbf{x}_{i_3} - \mathbf{x}_{i_4})\}^2 = Y_{2,n,p} - 2Y_{4,n,p} + Y_{5,n,p}. \end{aligned}$$

These two estimators also possess the location-invariance property. Based on $T_{1,n,p}$ and $T_{2,n,p}$, Chen et al. (2010) proposed the following test statistics for the identity hypothesis (1) and the sphericity hypothesis (2)

$$V_{n,p}^{CZZ} = (n/2)\{(T_{2,n,p} - 2T_{1,n,p})/p + 1\}, \quad (9)$$

$$U_{n,p}^{CZZ} = (n/2)(pT_{2,n,p}/T_{1,n,p}^2 - 1), \quad (10)$$

in their modification of the classical test statistics in (3) and (4).

For the identity testing problem, Ahmad (2016) considered the same choice of test statistic as in (9). We further let

$$\tilde{Y}_{2,n,p} = (n)_2^{-1} \sum_{(i_1, i_2)}^n \mathbf{x}_{i_1}^T \mathbf{x}_{i_1} \mathbf{x}_{i_2}^T \mathbf{x}_{i_2}, \quad \tilde{Y}_{4,n,p} = (n)_3^{-1} \sum_{(i_1, i_2, i_3)}^n \mathbf{x}_{i_1}^T \mathbf{x}_{i_1} \mathbf{x}_{i_2}^T \mathbf{x}_{i_2} \mathbf{x}_{i_3}^T \mathbf{x}_{i_3}, \quad \tilde{Y}_{5,n,p} = Y_{5,n,p}.$$

For the sphericity testing problem, Ahmad (2016) does not use $T_{1,n,p}^2$ as a plug-in estimator of $\text{tr}^2(\Sigma)$, but rather define its direct estimator as

$$\begin{aligned} T_{3,n,p} &= \{4(n)_4\}^{-1} \sum_{(i_1, i_2, i_3, i_4)}^n (\mathbf{x}_{i_1} - \mathbf{x}_{i_2})^T (\mathbf{x}_{i_1} - \mathbf{x}_{i_2}) (\mathbf{x}_{i_3} - \mathbf{x}_{i_4})^T (\mathbf{x}_{i_3} - \mathbf{x}_{i_4}), \\ &= \tilde{Y}_{2,n,p} - 2\tilde{Y}_{4,n,p} + \tilde{Y}_{5,n,p}. \end{aligned}$$

See Himeno and Yamada (2014) for more details and the properties of $T_{j,n,p}$, $j = 1, 2, 3$ for observations from the data structure (5). The test statistic of Ahmad (2016) for the sphericity hypothesis (2) is

$$U_{n,p}^A = (n/2)(pT_{2,n,p}/T_{3,n,p} - 1). \quad (11)$$

Similar to the classical test statistics in (3) and (4), $V_{n,p}^{CZZ}$, $U_{n,p}^{CZZ}$ and $U_{n,p}^A$ can be understood from the scaled distance measures of

$$\tau_1^2 = \text{tr}\{(\Sigma - \mathbf{I}_p)^2\}/p, \quad \text{and} \quad \tau_2^2 = \text{tr}\{[\Sigma/\{(1/p)\text{tr}(\Sigma)\} - \mathbf{I}_p]^2\}/p,$$

even if the population follows an elliptical structure. By virtue of the following Lemma 1, checking this intuition is straightforward.

Lemma 1. *Regardless of the independent component structure (5) and the elliptical model (7), $T_{1,n,p}$, $T_{2,n,p}$ and $T_{3,n,p}$ are the unbiased estimators of $\text{tr}(\Sigma)$, $\text{tr}(\Sigma^2)$ and $\text{tr}^2(\Sigma)$, respectively.*

The distinctive property for $V_{n,p}^{CZZ}$, $U_{n,p}^{CZZ}$ and $U_{n,p}^A$ is that their null asymptotic normality results are universal without specifying an explicit relationship between p and n as long as $p \rightarrow \infty$ as $n \rightarrow \infty$. Moreover, as shown by Cai and Ma (2013) under Gaussian assumption, they are rate optimal. However, the asymptotic validity depends on the data structure (5). Zou et al. (2014) used the numerical evidence to show this fact, and found it would be difficult to theoretically analyze the failure reason of sphericity test by Chen et al. (2010) under the general elliptical model in (7). To fill up this gap and get a broader picture of the U-tests by Chen et al. (2010) and Ahmad (2016), we then give new theoretical perspectives on their asymptotic behaviors when the underlying distribution belongs to the elliptical family. Unlike the pseudo-independent component model, the dependence between entries of the elliptically distributed data cannot be broken up in a linear fashion, complicating the theoretical analysis.

2.2 | Asymptotic normality under elliptical distributions

To derive the asymptotic property under elliptical distributions, we impose the following assumptions.

Assumption 1. As $n \rightarrow \infty$, $p = p(n) \rightarrow \infty$.

Assumption 2. An elliptically distributed random vector $\mathbf{x}_i$ has the stochastic representation (7) with the random radius R_i satisfying

$$E(R_i^2) = p, \quad E(R_i^4) = \varpi p^2 + \tau p + o(p), \quad E|\{R_i^2 - E(R_i^2)\}/\text{var}^{1/2}(R_i^2)|^{2+\varepsilon} < \infty,$$

for constants $\varpi \geq 1$, $\tau \geq 0$ and $\varepsilon > 0$.

In Assumption 1, we do not assume any explicit relationship between p and n , in the sense that $p(n)$ may go to infinity in a totally arbitrary way.

Assumption 2 characterizes the class of elliptical distributions. A specific elliptical distribution can be recognized by evaluating ϖ and τ . We verify this assumption for five commonly

TABLE 1 Some elliptical distributions and their limiting ratios.

$\mathbf{x}_i = \Lambda R_i \mathbf{u}_i + \mathbf{u}$	Distribution of R_i	ϖ	τ
Normal	$R_i^2 \sim \chi_p^2$	1	2
Double exponential	$R_i^2 \sim \text{Gamma}(p, 1)$	1	4
Exponential power	$R_i^{2s} \sim \text{Gamma}(2^{-1}s^{-1}p, 2^{-1}), s > 0$	1	$2/s$
Student- t	$R_i^2/p \sim F(p, v), v > 4$	$1 + \frac{2}{v-4}$	$\frac{2(v-2)}{v-4}$
Normal scale mixture	$R_i^2 = w_i v_i, w_i \perp v_i, v_i \sim \chi_p^2$	$1 + \frac{\text{var}(w_i)}{E^2(w_i)}$	$\frac{2E(w_i^2)}{E^2(w_i)}$

Note: The notation “ $\perp$ ” in the last row denotes independence.

used elliptical distributions in Table 1. Compared with Condition (2.1) in Hu et al. (2019) and Assumption A_E in Yin et al. (2023), it excludes the multivariate t distribution and mixtures of multivariate normal distributions whereas our Assumption 2 holds for these two families of distributions. This implies we focus on the tests that can deal with the distributions whose sample eigenvalue do not obey the generalized Marčenko–Pastur law (Li & Yao, 2018). From the Lyapunov-type condition in Assumption 2, we may truncate a scalar variable R_i at certain positions. Note that the concentration property of quadratic forms regarding a random direction $\mathbf{u}_i$ has been obtained by Gao et al. (2017, Lemma 5). The truncation and concentration steps together yield the centralized inequalities related to certain quadratic forms for the stochastic expression $R_i \mathbf{u}_i$, which are needed to satisfy the assumptions of martingale central limit theorem.

We consider $\sigma_{0,n,p}^2$ to quantify the asymptotic null variance of $V_{n,p}^{CZZ}$, $U_{n,p}^{CZZ}$, and $U_{n,p}^A$ under elliptical distributions, which is defined as

$$\begin{aligned} \sigma_{0,n,p}^2 = & 2n^{-2}p^2\{3p^2(p+2)^{-2}E^2(R_1^4)E^{-4}(R_1^2) - 1\} \\ & - 4n^{-2}p^2\{p(p+2)^{-1}E(R_1^4)E^{-2}(R_1^2) - 1\}. \end{aligned} \quad (12)$$

Since the right-hand side of (12) can be written as $4n^{-2}p^4(p+2)^{-2}E^2(R_1^4)E^{-4}(R_1^2) + 2n^{-2}p^2\{p(p+2)^{-1}E(R_1^4)E^{-2}(R_1^2) - 1\}^2$, we have $\sigma_{0,n,p}^2$ is always positive.

The following theorem establishes the asymptotic null distributions of $V_{n,p}^{CZZ}$, $U_{n,p}^{CZZ}$, and $U_{n,p}^A$ for the family of elliptical distributions.

Theorem 1. Suppose that Assumptions 1 and 2 hold for the elliptical model (7). Under the null identity hypothesis in (1),

$$2\sigma_{0,n,p}^{-1}(p/n) V_{n,p}^{CZZ} \xrightarrow{D} N(0, 1).$$

And under the null sphericity hypothesis in (2),

$$2\sigma_{0,n,p}^{-1}(p/n) U_{n,p}^{CZZ} \xrightarrow{D} N(0, 1),$$

$$2\sigma_{0,n,p}^{-1}(p/n) U_{n,p}^A \xrightarrow{D} N(0, 1),$$

where $\xrightarrow{D}$ denotes convergence in distribution.

As $E(R_1^4)/E^2(R_1^2) = \varpi + \tau/p + o(p^{-1})$, we note that the squared ratio has the following expression

$$\begin{aligned} \left\{ 2\sigma_{0,n,p}^{-1} (p/n) \right\}^2 &= 2^{-1} \{ 3p^2(p+2)^{-2} E^2(R_1^4) E^{-4}(R_1^2) - 2p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) + 1 \} \\ &= p^2(p+2)^{-2} E^2(R_1^4) E^{-4}(R_1^2) + 2^{-1} \{ p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) - 1 \}^2 \\ &\rightarrow \varpi^2 + 2^{-1}(\varpi - 1)^2 \\ &\geq 1, \end{aligned} \quad (13)$$

where the last inequality follows from Hölder's inequality. Thus, we have $2\sigma_{0,n,p}^{-1} (p/n) \geq 1$ as $n, p \rightarrow \infty$. The relation (13) readily implies that the limiting laws in Theorem 1 have different forms for $\varpi = 1$ and $\varpi > 1$. If $\varpi = 1$, the identity and sphericity U-tests introduced by Chen et al. (2010) and Ahmad (2016) keep asymptotically valid. However, for the elliptical distributions with $\varpi > 1$, the location-invariant tests of Chen et al. (2010) and Ahmad (2016) would have much larger type I error rates than the nominal levels, which echoes Zou et al. (2014, Tbl. 2). This is mainly due to the strong dependence between the coordinates of elliptically high-dimensional data. Motivated by this phenomenon, we develop new test statistics for testing (1) and (2) in the high dimensions.

3 | THE ADJUSTED TEST STATISTICS

To deal with the elliptical populations with $\varpi > 1$, we need to provide durable and accurate estimator for $E(R_1^4)/E^2(R_1^2)$ in the asymptotic variance $\sigma_{0,n,p}^2$. The key idea of our proposal is to estimate $\text{tr}^2(\Sigma)E(R_1^4)/E^2(R_1^2)$ and $\text{tr}^2(\Sigma)$, respectively. Then, their ratio yields the estimator for $E(R_1^4)/E^2(R_1^2)$. We further define

$$Y_{6,n,p} = (n)_1^{-1} \sum_{i_1=1}^n (\mathbf{x}_{i_1}^T \mathbf{x}_{i_1})^2, \quad Y_{7,n,p} = (n)_2^{-1} \sum_{(i_1, i_2)}^n \mathbf{x}_{i_1}^T \mathbf{x}_{i_1} \mathbf{x}_{i_1}^T \mathbf{x}_{i_2}, \quad Y_{8,n,p} = \tilde{Y}_{4,n,p}.$$

The estimator of $\text{tr}^2(\Sigma)E(R_1^4)/E^2(R_1^2)$ is given by

$$\begin{aligned} \delta_{n,p} &= (n)_5^{-1} \sum_{(i_1, i_2, i_3, i_4, i_5)}^n (\mathbf{x}_{i_1} - \mathbf{x}_{i_2})^T (\mathbf{x}_{i_1} - \mathbf{x}_{i_3}) (\mathbf{x}_{i_1} - \mathbf{x}_{i_4})^T (\mathbf{x}_{i_1} - \mathbf{x}_{i_5}) \\ &= Y_{6,n,p} - 4Y_{7,n,p} + 2Y_{8,n,p} + 4Y_{4,n,p} - 3Y_{5,n,p}, \end{aligned}$$

whose unbiasedness will be shown in Theorem 2. The term $\delta_{n,p}$ is the U-statistic of order five, and remains invariant after the location transformation.

Based on $\delta_{n,p}$ and $T_{3,n,p}$, a direct estimator for $\sigma_{0,n,p}^2$ is

$$\begin{aligned} \hat{\sigma}_{0,n,p}^2 &= 2n^{-2} p^2 \{ 3p^2(p+2)^{-2} \delta_{n,p}^2 / T_{3,n,p}^2 - 1 \} \\ &\quad - 4n^{-2} p^2 \{ p(p+2)^{-1} \delta_{n,p} / T_{3,n,p} - 1 \}. \end{aligned} \quad (14)$$

The right-hand side of (14) equals $4n^{-2} p^4 (p+2)^{-2} \delta_{n,p}^2 / T_{3,n,p}^2 + 2n^{-2} p^2 \{ p^{-1} (p+2)^{-1} \delta_{n,p} / T_{3,n,p} - 1 \}^2$, which guarantees that $\hat{\sigma}_{0,n,p}^2$ is always positive. Then, the adjusted test statistics we use for testing

the identity hypothesis (1) and the sphericity hypothesis (2) are

$$\begin{aligned}\hat{V}_{n,p} &= \hat{\sigma}_{0,n,p}^{-1} p \{ (T_{2,n,p} - 2T_{1,n,p})/p + 1 \}, \\ \hat{U}_{n,p} &= \hat{\sigma}_{0,n,p}^{-1} p (pT_{2,n,p}/T_{3,n,p} - 1).\end{aligned}$$

Remark 1. To estimate $\text{tr}^2(\Sigma)$, an alternative way is using $T_{1,n,p}^2$, which is the square estimator of $\text{tr}(\Sigma)$ introduced by Chen et al. (2010). We carefully compare two estimators $T_{1,n,p}^2$ and $T_{3,n,p}$, and found

$$\begin{aligned}T_{1,n,p}^2 &= \{(n-2)(n-3)/(n)_2\} T_{3,n,p} \\ &+ (n)_2^{-2} \sum_{(i_1, i_2, i_3)}^n (\mathbf{x}_{i_1} - \mathbf{x}_{i_2})^T (\mathbf{x}_{i_1} - \mathbf{x}_{i_2}) (\mathbf{x}_{i_1} - \mathbf{x}_{i_3})^T (\mathbf{x}_{i_1} - \mathbf{x}_{i_3}) \\ &+ \{2(n)_2^2\}^{-1} \sum_{(i_1, i_2)}^n \{(\mathbf{x}_{i_1} - \mathbf{x}_{i_2})^T (\mathbf{x}_{i_1} - \mathbf{x}_{i_2})\}^2.\end{aligned}$$

In fact, only the term $\{(n-2)(n-3)/(n)_2\} T_{3,n,p}$ contains the summation of cross product terms of mutually different $\mathbf{x}_{i_1}$, $\mathbf{x}_{i_2}$, $\mathbf{x}_{i_3}$ and $\mathbf{x}_{i_4}$, which is useful for testing. The remaining two terms in $T_{1,n,p}^2$ would introduce bias terms in the test. Therefore, following Ahmad (2016), we suggest using $T_{3,n,p}$ as the estimator of $\text{tr}^2(\Sigma)$.

Remark 2. In the high-dimensional setting, computation cost is also worthy of consideration. Since the estimators involve the high-order U-statistics, direct implementations of $\hat{V}_{n,p}$ and $\hat{U}_{n,p}$ have computation complexity of $O(n^5)$. To reduce the computational burden, we design a computationally efficient algorithm, which is given in the Section S7. Through this algorithm, we need no more than $O(n^2)$ steps to calculate both $\hat{V}_{n,p}$ and $\hat{U}_{n,p}$.

4 | MAIN RESULTS

In this section, we study the asymptotic distributions of the new test statistics $\hat{V}_{n,p}$ and $\hat{U}_{n,p}$ under two different kinds of nonparametric distribution assumptions: (I) the elliptical distributions (7); (II) the pseudo-independence models (5). Our proposed tests allow for universal (n, p) -asymptotics, in the sense that the respective null asymptotic distributions hold whenever $\min(n, p) \rightarrow \infty$.

For the later exploration of power, we need one more assumption:

Assumption 3. $\text{tr}(\Sigma^2) \rightarrow \infty$ and $\text{tr}(\Sigma^4)/\text{tr}^2(\Sigma^2) \rightarrow 0$, as $p \rightarrow \infty$.

If the null hypothesis H_0 or $\tilde{H}_0$ holds, Assumption 3 is trivially satisfied. This assumption is also used by Chen et al. (2010), Chen and Qin (2010), Zhong and Chen (2011), Wang et al. (2015), Ahmad (2016) and Cui et al. (2018) to derive the asymptotic properties of their test statistics. They revealed that Assumption 3 always holds for the covariance Σ , whose eigenvalues are bounded away from zero and infinity. Moreover, even if some of eigenvalues are unbounded, Chen and Qin (2010) showed that Assumption 3 may still hold. Some commonly encountered covariance structures such as autoregression, banded structures meet this assumption.

4.1 | Results on elliptical distribution assumption

We start with notations which quantify the asymptotic variances of $\hat{V}_{n,p}$ and $\hat{U}_{n,p}$ for elliptical distributions. Let

$$\begin{aligned}\sigma_{1,n,p}^2 &= 2n^{-2} \{ 3p^2(p+2)^{-2} E^2(R_1^4) E^{-4}(R_1^2) - 1 \} \text{tr}^2(\Sigma^2) \\ &\quad - 4n^{-2} \{ p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) - 1 \} \text{tr}^2(\Sigma^2) \\ &\quad + 8n^{-1} p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) \text{tr}(\Sigma^2 - \Sigma)^2 \\ &\quad + 4n^{-1} \{ p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) - 1 \} \\ &\quad \times \text{tr}^2(\Sigma^2 - \Sigma), \\ \sigma_{2,n,p}^2 &= 2n^{-2} \{ 3p^2(p+2)^{-2} E^2(R_1^4) E^{-4}(R_1^2) - 1 \} \\ &\quad - 4n^{-2} \{ p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) - 1 \} \\ &\quad + 8n^{-1} p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) \\ &\quad \times \text{tr} \{ \Sigma^2 / \text{tr}(\Sigma^2) - \Sigma / \text{tr}(\Sigma) \}^2.\end{aligned}$$

By applying Hölder's inequality, $\text{tr}^2(\Sigma^2 - \Sigma) \leq p \text{tr}(\Sigma^2 - \Sigma)^2$, $\text{tr}^2 \{ \Sigma^2 / \text{tr}(\Sigma^2) - \Sigma / \text{tr}(\Sigma) \} \leq p \text{tr} \{ \Sigma^2 / \text{tr}(\Sigma^2) - \Sigma / \text{tr}(\Sigma) \}^2$ and $E^2(R_1^2) \leq E(R_1^4)$. These ensure that

$$\begin{aligned}\sigma_{1,n,p}^2 &\geq 2n^{-2} \{ 3p^2(p+2)^{-2} E^2(R_1^4) E^{-4}(R_1^2) - 1 \} \text{tr}^2(\Sigma^2) \\ &\quad - 4n^{-2} \{ p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) - 1 \} \text{tr}^2(\Sigma^2) > 0 \\ \sigma_{2,n,p}^2 &\geq 2n^{-2} \{ 3p^2(p+2)^{-2} E^2(R_1^4) E^{-4}(R_1^2) - 1 \} \\ &\quad - 4n^{-2} \{ p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) - 1 \} > 0.\end{aligned}$$

If $\Sigma = \mathbf{I}_p$, $\sigma_{1,n,p}^2$ reduces to $\sigma_{0,n,p}^2$. When $\Sigma = \sigma^2 \mathbf{I}_p$, $\sigma_{2,n,p}^2$ becomes $p^{-2} \sigma_{0,n,p}^2$.

The theoretical properties of $\hat{V}_{n,p}$, $\hat{U}_{n,p}$ and $\delta_{n,p}$ under the elliptical distribution assumption are presented in the following theorem.

Theorem 2. Suppose that Assumptions 1 and 2 hold for model (7).

- (i) The expectation of $\delta_{n,p}$ is $E(\delta_{n,p}) = p(p+2)^{-1} E(R_1^4) E^{-2}(R_1^2) \{ \text{tr}^2(\Sigma) + 2 \text{tr}(\Sigma^2) \}$. In particular, under the null hypothesis in either (1) or (2), $E(\delta_{n,p}) = \text{tr}^2(\Sigma) E(R_1^4) / E^2(R_1^2)$.
- (ii) Under Assumption 3, $\hat{\sigma}_{0,n,p}^2$ is ratio-consistent to

$$2n^{-2} p^2 \{ 3p^2(p+2)^{-2} E^2(\delta_{n,p}) / \text{tr}^4(\Sigma) - 1 \} - 4n^{-2} p^2 \{ p(p+2)^{-1} E(\delta_{n,p}) / \text{tr}^2(\Sigma) - 1 \}.$$

In particular, under the null hypothesis in either (1) or (2), $\hat{\sigma}_{0,n,p}^2$ is ratio-consistent to $\sigma_{0,n,p}^2$.

With the aid of Theorem 2, the martingale central limit theorem (Hall & Heyde, 2014) and Slutsky's theorem, we establish the asymptotic distributions of $\hat{V}_{n,p}$ and $\hat{U}_{n,p}$ in the following theorem.

Theorem 3. Suppose that Assumptions 1–3 hold for elliptical distributions (7),

$$\begin{aligned}\sigma_{1,n,p}^{-1} p (\sigma_{0,n,p} \hat{V}_{n,p}/p - \tau_1^2) &\xrightarrow{D} N(0, 1), \\ \sigma_{2,n,p}^{-1} \{p^{-1} \text{tr}^2(\Sigma)/\text{tr}(\Sigma^2)\} (\sigma_{0,n,p} \hat{U}_{n,p}/p - \tau_2^2) &\xrightarrow{D} N(0, 1).\end{aligned}$$

In particular, under the null hypothesis in (1), $\hat{V}_{n,p} \xrightarrow{D} N(0, 1)$, and under the null hypothesis in (2), $\hat{U}_{n,p} \xrightarrow{D} N(0, 1)$.

Let z_α be the upper α quantile of $N(0, 1)$. From Theorem 3, our α -level tests for the identity and sphericity hypotheses have the same rejection regions, that is,

$$\hat{V}_{n,p} \geq z_\alpha, \quad \text{and} \quad \hat{U}_{n,p} \geq z_\alpha.$$

We then turn to evaluate the powers of our identity and sphericity tests under elliptical distributions. Let $\beta_{1,n,p} = P(\hat{V}_{n,p} \geq z_\alpha | \Sigma \neq \mathbf{I}_p)$ be the power of the proposed identity test under $H_1 : \Sigma \neq \mathbf{I}_p$. Similarly, let $\beta_{2,n,p} = P(\hat{U}_{n,p} \geq z_\alpha | \Sigma \neq \sigma^2 \mathbf{I}_p)$ be the power of the proposed sphericity test under $\tilde{H}_1 : \Sigma \neq \sigma^2 \mathbf{I}_p$. Based on Theorems 2 and 3, the proposed identity test has the asymptotic power

$$\beta_{1,n,p} = \{1 + o(1)\} \Phi(-B_{1,n,p} z_\alpha + p \tau_1^2 / \sigma_{1,n,p}), \quad (15)$$

where $\Phi(\cdot)$ is the standard normal distribution function and $B_{1,n,p}$ is the nonnegative number given by

$$\begin{aligned}B_{1,n,p}^2 &= \sigma_{1,n,p}^{-2} [2n^{-2} p^2 \{3p^2(p+2)^{-2} E^2(\delta_{n,p})/\text{tr}^4(\Sigma) - 1\} \\ &\quad - 4n^{-2} p^2 \{p(p+2)^{-1} E(\delta_{n,p})/\text{tr}^2(\Sigma) - 1\}].\end{aligned}$$

The proposed sphericity test has the asymptotic power

$$\beta_{2,n,p} = \{1 + o(1)\} \Phi[-B_{2,n,p} z_\alpha + \{p^{-1} \text{tr}^2(\Sigma)/\text{tr}(\Sigma^2)\} \tau_2^2 / \sigma_{2,n,p}], \quad (16)$$

where $B_{2,n,p}$ is the nonnegative number defined as

$$B_{2,n,p}^2 = (\sigma_{1,n,p}^2 / \sigma_{2,n,p}^2) \{p^{-1} \text{tr}^2(\Sigma)/\text{tr}(\Sigma^2)\}^2 B_{1,n,p}^2.$$

We study the consistency of two proposed tests for elliptically distributed data in the next theorem.

Theorem 4. Suppose that Assumptions 1–3 hold for elliptical distributions (7),

- (i) If $n\tau_1^2 \rightarrow \infty$, then $B_{1,n,p}$ is bounded, $p\tau_1^2/\sigma_{1,n,p} \rightarrow \infty$ and $\beta_{1,n,p} \rightarrow 1$.
- (ii) If $n\tau_2^2 \rightarrow \infty$, then $B_{2,n,p}$ is bounded, $\{p^{-1} \text{tr}^2(\Sigma)/\text{tr}(\Sigma^2)\} \tau_2^2 / \sigma_{2,n,p} \rightarrow \infty$ and $\beta_{2,n,p} \rightarrow 1$.

Theorem 4 implies that if the difference between Σ and $\mathbf{I}_p$ is not too small so that τ_1^2 is at the same or a larger order of n^{-1} , the proposed identity test will be powerful. Similarly, the proposed sphericity test has nontrivial power as long as the measure of the departure τ_2^2 does not shrink

faster than n^{-1} . The assumptions $n\tau_1^2 \rightarrow \infty$ and $n\tau_2^2 \rightarrow \infty$ are trivially true for fixed-dimensional data as $n \rightarrow \infty$. For high-dimensional data, these two assumptions are less automatic as $\text{tr}(\Sigma^2)$ can diverge. We focus on the case that all the eigenvalues of Σ are bounded away from zero and infinity. The proposed tests are consistent if the average signals $p^{-1}\text{tr}^{1/2}\{(\Sigma - \mathbf{I}_p)^2\}$ and $p^{-1}\text{tr}^{1/2}[(\Sigma/\{(1/p)\text{tr}(\Sigma)\} - \mathbf{I}_p)^2]$ are at least at the order of $n^{-1/2}p^{-1/2}$, which is smaller than the conventional order of $n^{-1/2}$ for fixed-dimension situations. This phenomenon reflects the fact that high dimensionality is not entirely a curse as it provides more data information available as well.

4.2 | Adaptivity under pseudo-independence assumption

We have carefully studied the asymptotic behaviors for the family of elliptical distributions. It is possible to generalize our results to some other useful distributions beyond the elliptical family. We take the pseudo-independence component models in (5) as an example, and establish the asymptotic properties in what follows.

Theorem 5. *Suppose that Assumption 1 holds for pseudo-independence component models in (5). Under the null hypothesis in (1), $\hat{V}_{n,p} \xrightarrow{D} N(0, 1)$, and under the null hypothesis in (2), $\hat{U}_{n,p} \xrightarrow{D} N(0, 1)$.*

For pseudo-independence component data, Theorem 5 shows that $\hat{V}_{n,p}$ and $\hat{U}_{n,p}$ still have the asymptotic normality under hypotheses (1) and (2), without restricting the explicit growth rate of p relative to n . Recall the tests of Chen et al. (2010) and Ahmad (2016) are developed under the data structure (5), we then provide the limiting efficiency comparison with their tests in Theorem 6.

Theorem 6. *Suppose that Assumption 1 and 3 hold for pseudo-independence component models in (5). Then the proposed identity and sphericity tests are asymptotically as efficient as those of Chen et al. (2010) and Ahmad (2016).*

5 | NUMERICAL STUDIES

5.1 | Simulations

We conduct simulations to augment the universal asymptotic results. For the purposes of comparison, we consider the identity and sphericity tests of Chen et al. (2010) and Ahmad (2016), denoted by $T_{CZZ,V}$, $T_{CZZ,U}$, and $T_{A,U}$, multivariate sign test for sphericity with bias corrected by Zou et al. (2014, denoted by T_{BCS}). We also include the spatial sign tests for identity and sphericity (Hu et al., 2019; Yang et al., 2021), which require to estimate location parameters. In our simulations, we know in advance the true location parameters. Therefore, we use $T_{J, \text{true}}$ and $T_{J, \text{est}}$ to examine how the location estimates influence the performances of the tests of Hu et al. (2019) and Yang et al. (2021). The sample size n is 30 or 200, and the dimension p is 80, 200, and 2000. We evaluate the performances of the aforementioned tests in terms of empirical size and power. We use the significant level $\alpha = 5\%$ throughout our simulations and summarize the results based on $M = 5000$ replications.

For the sphericity test, we generate the random observations $\{\mathbf{x}_i, i = 1, \dots, n\}$ from six distributions. For the identity test, we standardize the shape matrix such that $\text{tr}(\Sigma) = p$. The

location parameters are all zero under the null hypotheses. The six distributions are as follows: (I) the multivariate standard normal distribution; (II) the pseudo-independence model with standardized $\chi^2(8)$ distribution; (III) the pseudo-independence model with standardized $t(8)$ distribution; (IV) the multivariate standard $t(8)$ distribution; (V) a mixture of two multivariate normal distributions $0.8\mathcal{N}(\mathbf{u}_{0,1}, \mathbf{I}_p) + 0.2\mathcal{N}(\mathbf{u}_{0,2}, 9\mathbf{I}_p)$; (VI) the multivariate standard $t(3)$ distribution. The mean vectors $\mathbf{u}_{0,1}$ and $\mathbf{u}_{0,2}$ are zeros unless stated otherwise. We remark here that, in scenario (I), the random vectors follow elliptical distribution and simultaneously satisfy the pseudo-independence model. In scenarios (II) and (III), the random vectors are not elliptically distributed but satisfy the pseudo-independence model. In scenarios (IV) and (V), the random vectors are elliptically distributed but do not satisfy the pseudo-independence model. The uniformly bounded eighth moments are all satisfied by scenarios (I), (II), (III), (IV), and (V). We also consider a scenario (VI) where such constraints on the moments are violated.

To evaluate the power performance, we generate the random vectors $\mathbf{y}_i$ to obtain $\mathbf{x}_i = \mathbf{A}\mathbf{y}_i$. In the first four scenarios, we set $\mathbf{A} = \text{diag}(2^{1/2}\mathbf{1}_{[vp]}, \mathbf{1}_{p-[vp]})$, where $[vp]$ denotes the integer truncation of vp , $\mathbf{1}_{[vp]} \in \mathbb{R}^{[vp]}$ is the vector of all ones, and $\text{diag}(\cdot)$ denotes the diagonalization of a matrix. We use $v = 0.2$ for $n = 30$ and $v = 0.03$ for $n = 200$. In the fifth scenario, we choose $\mathbf{A}^2$ to be the matrix whose diagonal elements are 1 and off-diagonal ones are ρ , where $\rho = 0.01$ for $n = 30$ and $\rho = 0.001$ for $n = 200$. The elements in $\mathbf{u}_{0,1}$ are set to be 0.5, and $\mathbf{u}_{0,2}$ is still a vector of zeros. For the last scenario, we consider $\mathbf{A} = \{(1 - \rho)\mathbf{I}_p + \rho\mathbf{1}_p\mathbf{1}_p^T\}^{1/2} = (1 - \rho)^{1/2}\mathbf{I}_p + p^{-1}\{(1 - \rho + p\rho)^{1/2} - (1 - \rho)^{1/2}\}\mathbf{1}_p\mathbf{1}_p^T$, where we let $\rho = 0.05$ for $n = 30$ and $\rho = 0.015$ for $n = 200$.

The empirical sizes of all the tests are summarized in Table 2 for all scenarios. From Table 2, the empirical sizes of our proposed tests are quite reasonably around the level 0.05, across all simulated distributions. The empirical distributions of the proposed test statistics are clearly compatible with Theorems 3 and 5. For both statistics $\hat{V}_{n,p}$ and $\hat{U}_{n,p}$, the Gaussian approximation is valid for the moderate to large values of n and p , irrespective of the ratio p/n . The size performances imply the proposed tests are applicable to the data are generated from normal scenario (I), pseudo-independence scenarios (II) and (III), and elliptical scenarios (IV) and (V), which also confirm our universal asymptotic results. Under the scenario (VI), the proposed tests still maintain reasonable type I error rates, even though the moment constraints are not satisfied.

For the identity and sphericity tests of Chen et al. (2010) and Ahmad (2016), they can achieve the nominal levels of size in a universal way under normal scenario (I), and pseudo-independence scenarios (II) and (III). However, for elliptical scenarios (IV), (V) and (VI), all tests based on $T_{CZZ,V}$, $T_{CZZ,U}$ and $T_{A,U}$ are not able to control their empirical sizes very well. The considerable bias terms in test statistics make them have type I error rates that are much larger than 0.05. This phenomena are consistent with the conclusions in Theorem 1.

The empirical sizes of the bias-corrected sign test T_{BCS} have satisfactory performances for normal scenario (I), and pseudo-independence scenarios (II) and (III). As mentioned in Zou et al. (2014), the validity of their test is established under $p = O(n^2)$. For $n = 30$ and $p = 2000$, we see that T_{BCS} has fragile size performances under elliptical scenarios (IV), (V) and (VI). This issue may prevent T_{BCS} from applied to the situation where p grows faster than n^2 .

With the known location parameters, the spatial sign test $T_{\text{J,true}}$ maintains the reasonable empirical sizes which are close to 0.05. However, when the location parameters are substituted by the estimator $n^{-1}\sum_{i=1}^n \mathbf{x}_i$, the empirical sizes of $T_{\text{J,est}}$ inflate quickly as p increases, even reaching one for $p = 2000$. This is mainly due to the bias terms arising from the location estimation. By contrast, our proposed tests benefit from the location-invariant property.

TABLE 2 The empirical sizes for scenarios (I)–(VI) with several combinations of (n, p) when $\alpha = 0.05$.

		(n, p)					
	Method	(30,80)	(30,200)	(30,2000)	(200,80)	(200,200)	(200,2000)
S(I)	$T_{\hat{V}_{n,p}}$	0.063	0.055	0.064	0.056	0.052	0.047
	$T_{\hat{U}_{n,p}}$	0.063	0.055	0.064	0.056	0.053	0.047
	$T_{CZZ,V}$	0.062	0.055	0.064	0.057	0.053	0.047
	$T_{CZZ,U}$	0.062	0.054	0.064	0.053	0.052	0.047
	$T_{A,U}$	0.059	0.063	0.062	0.055	0.053	0.056
	T_{BCS}	0.055	0.048	0.066	0.050	0.053	0.046
	$T_{J, \text{true}}$	0.048	0.046	0.052	0.049	0.050	0.047
	$T_{J, \text{est}}$	0.366	0.966	1.000	0.075	0.124	1.000
S(II)	$T_{\hat{V}_{n,p}}$	0.068	0.068	0.061	0.053	0.058	0.052
	$T_{\hat{U}_{n,p}}$	0.069	0.069	0.061	0.053	0.057	0.052
	$T_{CZZ,V}$	0.072	0.069	0.061	0.056	0.060	0.052
	$T_{CZZ,U}$	0.070	0.069	0.061	0.054	0.058	0.052
	$T_{A,U}$	0.073	0.068	0.060	0.049	0.051	0.061
	T_{BCS}	0.058	0.058	0.064	0.052	0.057	0.050
	$T_{J, \text{true}}$	0.049	0.053	0.051	0.047	0.054	0.048
	$T_{J, \text{est}}$	0.373	0.965	1.000	0.069	0.130	1.000
S(III)	$T_{\hat{V}_{n,p}}$	0.067	0.062	0.067	0.061	0.059	0.057
	$T_{\hat{U}_{n,p}}$	0.065	0.063	0.066	0.060	0.058	0.056
	$T_{CZZ,V}$	0.071	0.063	0.067	0.063	0.061	0.058
	$T_{CZZ,U}$	0.065	0.063	0.066	0.061	0.058	0.056
	$T_{A,U}$	0.063	0.071	0.062	0.066	0.050	0.046
	T_{BCS}	0.054	0.057	0.066	0.056	0.055	0.056
	$T_{J, \text{true}}$	0.049	0.051	0.053	0.054	0.052	0.054
	$T_{J, \text{est}}$	0.371	0.960	1.000	0.077	0.132	1.000
S(IV)	$T_{\hat{V}_{n,p}}$	0.065	0.052	0.057	0.057	0.051	0.051
	$T_{\hat{U}_{n,p}}$	0.059	0.050	0.056	0.052	0.048	0.046
	$T_{CZZ,V}$	0.130	0.111	0.113	0.136	0.141	0.134
	$T_{CZZ,U}$	0.102	0.091	0.094	0.101	0.104	0.101
	$T_{A,U}$	0.124	0.120	0.123	0.126	0.132	0.122
	T_{BCS}	0.058	0.061	0.077	0.057	0.043	0.048
	$T_{J, \text{true}}$	0.054	0.053	0.053	0.051	0.041	0.047
	$T_{J, \text{est}}$	0.419	0.981	1.000	0.080	0.133	1.000

TABLE 2 (Continued)

		(n, p)					
	Method	(30,80)	(30,200)	(30,2000)	(200,80)	(200,200)	(200,2000)
S(V)	$T_{\hat{V}_{n,p}}$	0.061	0.064	0.057	0.058	0.058	0.053
	$T_{\hat{U}_{n,p}}$	0.043	0.046	0.045	0.040	0.044	0.040
	$T_{CZZ,V}$	0.233	0.248	0.226	0.260	0.269	0.252
	$T_{CZZ,U}$	0.139	0.152	0.136	0.167	0.172	0.164
	$T_{A,U}$	0.131	0.132	0.128	0.159	0.154	0.136
	T_{BCS}	0.054	0.056	0.085	0.047	0.054	0.052
	$T_{J, \text{true}}$	0.048	0.050	0.045	0.114	0.055	0.051
	$T_{J, \text{est}}$	0.804	1.000	1.000	0.046	0.309	1.000
S(VI)	$T_{\hat{V}_{n,p}}$	0.053	0.060	0.048	0.059	0.057	0.061
	$T_{\hat{U}_{n,p}}$	0.036	0.042	0.044	0.042	0.046	0.043
	$T_{CZZ,V}$	0.209	0.220	0.198	0.287	0.310	0.272
	$T_{CZZ,U}$	0.106	0.127	0.119	0.151	0.170	0.148
	$T_{A,U}$	0.181	0.190	0.184	0.249	0.265	0.216
	T_{BCS}	0.054	0.072	0.079	0.053	0.043	0.058
	$T_{J, \text{true}}$	0.045	0.069	0.057	0.049	0.043	0.049
	$T_{J, \text{est}}$	0.672	0.997	1.000	0.105	0.245	1.000

Table 3 reports the empirical powers of all the tests. In terms of the performances under alternatives, the proposed tests have comparable powers with the tests of Chen et al. (2010) and Ahmad (2016), and the test of Zou et al. (2014) when the empirical sizes are well controlled. This is consistent with the conclusions drawn from Theorem 6. Thanks to the blessing of dimensionality, the powers of our proposed test increase gradually as p increases. Under some elliptical scenarios, the severely distorted sizes for $T_{CZZ,V}$, $T_{CZZ,U}$, $T_{A,U}$, and $T_{J, \text{est}}$ also make their powers artificially high.

5.2 | A gene expression data application

In this section, we consider a dataset recorded by Khan et al. (2001). They performed a small round blue-cell tumors micro-array experiment, and obtained $n = 64$ training tissue samples with 6567 gene expression values for each sample. The gene expressions were then filtered down, with $p = 2308$ values remained. There are four types of tumors in the training tissue samples: 23 Ewing family of tumors (EWS), 8 Burkitt lymphoma, a subset of non-Hodgkin lymphoma (BL-NHL), 12 from neuroblastoma (NB), and 21 from rhabdomyosarcoma (RMS). This dataset has been analyzed by Rothman et al. (2009), Cai and Liu (2011), and Xue et al. (2012) to obtain sparse estimates of the covariance matrix. We test whether the covariance matrix is spherical.

We centered the gene expressions within each group, and apply $T_{\hat{U}_{n,p}}$, $T_{CZZ,U}$, $T_{A,U}$, T_{BCS} , and $T_{J, \text{est}}$ to examine the spherical hypothesis. The p -values of these four tests are all smaller than

TABLE 3 The empirical powers for scenarios (I)–(VI) with several combinations of (n, p) when $\alpha = 0.05$.

		(n, p)					
	Method	(30,80)	(30,200)	(30,2000)	(200,80)	(200,200)	(200,2000)
S(I)	$T_{\hat{V}_{n,p}}$	0.755	0.778	0.785	0.734	0.874	0.895
	$T_{\hat{U}_{n,p}}$	0.494	0.487	0.504	0.699	0.841	0.858
	$T_{CZZ,V}$	0.756	0.778	0.785	0.734	0.875	0.895
	$T_{CZZ,U}$	0.489	0.485	0.504	0.694	0.841	0.858
	$T_{A,U}$	0.481	0.473	0.490	0.698	0.829	0.845
	T_{BCS}	0.439	0.450	0.511	0.655	0.819	0.853
	$T_{J, \text{true}}$	0.435	0.453	0.480	0.644	0.817	0.858
	$T_{J, \text{est}}$	0.844	0.999	1.000	0.704	0.919	1.000
S(II)	$T_{\hat{V}_{n,p}}$	0.728	0.762	0.781	0.701	0.859	0.888
	$T_{\hat{U}_{n,p}}$	0.475	0.494	0.498	0.668	0.823	0.852
	$T_{CZZ,V}$	0.735	0.765	0.781	0.711	0.860	0.888
	$T_{CZZ,U}$	0.478	0.495	0.499	0.670	0.824	0.852
	$T_{A,U}$	0.438	0.486	0.493	0.655	0.808	0.839
	T_{BCS}	0.423	0.457	0.505	0.629	0.799	0.846
	$T_{J, \text{true}}$	0.422	0.452	0.476	0.616	0.798	0.849
	$T_{J, \text{est}}$	0.835	0.998	1.000	0.685	0.902	1.000
S(III)	$T_{\hat{V}_{n,p}}$	0.744	0.757	0.789	0.705	0.857	0.889
	$T_{\hat{U}_{n,p}}$	0.480	0.480	0.498	0.668	0.822	0.852
	$T_{CZZ,V}$	0.749	0.760	0.789	0.713	0.858	0.889
	$T_{CZZ,U}$	0.483	0.482	0.498	0.670	0.823	0.852
	$T_{A,U}$	0.477	0.481	0.483	0.669	0.805	0.839
	T_{BCS}	0.434	0.447	0.505	0.619	0.799	0.847
	$T_{J, \text{true}}$	0.437	0.448	0.478	0.614	0.796	0.849
	$T_{J, \text{est}}$	0.837	0.999	1.000	0.674	0.893	1.000
S(IV)	$T_{\hat{V}_{n,p}}$	0.526	0.535	0.525	0.477	0.599	0.610
	$T_{\hat{U}_{n,p}}$	0.324	0.318	0.311	0.448	0.558	0.569
	$T_{CZZ,V}$	0.635	0.653	0.641	0.653	0.765	0.782
	$T_{CZZ,U}$	0.420	0.426	0.415	0.574	0.700	0.714
	$T_{A,U}$	0.422	0.437	0.423	0.593	0.712	0.731
	T_{BCS}	0.438	0.462	0.531	0.660	0.825	0.849
	$T_{J, \text{true}}$	0.437	0.465	0.467	0.651	0.823	0.850
	$T_{J, \text{est}}$	0.871	1.000	1.000	0.717	0.922	1.000

TABLE 3 (Continued)

		(n, p)					
	Method	(30,80)	(30,200)	(30,2000)	(200,80)	(200,200)	(200,2000)
S(V)	$T_{\hat{V}_{n,p}}$	0.172	0.590	1.000	0.230	0.745	1.000
	$T_{\hat{U}_{n,p}}$	0.146	0.648	0.999	0.212	0.766	1.000
	$T_{CZZ,V}$	0.424	0.848	1.000	0.570	0.944	1.000
	$T_{CZZ,U}$	0.312	0.798	0.999	0.466	0.921	1.000
	$T_{A,U}$	0.310	0.786	1.000	0.463	0.917	1.000
	T_{BCS}	0.152	0.567	0.578	0.122	0.382	1.000
	$T_{J, \text{true}}$	0.256	0.948	1.000	0.371	0.975	1.000
	$T_{J, \text{est}}$	0.937	1.000	1.000	0.776	0.998	1.000
S(VI)	$T_{\hat{V}_{n,p}}$	0.539	0.765	0.908	0.551	0.763	1.000
	$T_{\hat{U}_{n,p}}$	0.506	0.707	0.902	0.526	0.774	1.000
	$T_{CZZ,V}$	0.687	0.875	0.984	0.625	0.884	1.000
	$T_{CZZ,U}$	0.541	0.783	0.975	0.599	0.796	1.000
	$T_{A,U}$	0.662	0.890	0.990	0.597	0.895	1.000
	T_{BCS}	0.603	0.811	0.961	0.686	0.904	1.000
	$T_{J, \text{true}}$	0.659	0.839	1.000	0.684	0.903	1.000
	$T_{J, \text{est}}$	0.981	1.000	1.000	0.813	1.000	1.000

TABLE 4 The rejection probabilities for seven tests when $\alpha = 0.05$.

Method	$T_{\hat{V}_{n,p}}$	$T_{\hat{U}_{n,p}}$	$T_{CZZ,V}$	$T_{CZZ,U}$	$T_{A,U}$	T_{BCS}	$T_{J, \text{est}}$
	0.055	0.062	0.077	0.077	0.075	0.170	0.340

10^{-10} , which strongly supports the fact that the covariance matrix for gene expressions is not spherical.

Furthermore, we follow Rothman et al. (2009) to reorder the gene expressions by their corresponding F -statistics, divide the total 2308 genes by 23 groups based on the orders. The first 22 groups each contain 100 genes, and the last one has 108 genes. Previous studies show that the dependencies between the genes belonging to different groups are weak, resulting in the sparse covariance matrix. Thus, we are motivated to conduct the identity and sphericity tests to examine the group-wise dependence. For each group, we choose i th position gene for $i = 1, \dots, 100$, to make up the targeted $\mathbf{x}$ of 23 cross-group genes. We obtain 100 selected gene sets in total. Each gene set is standardized, and then is performed the seven tests $T_{\hat{V}_{n,p}}, T_{\hat{U}_{n,p}}, T_{CZZ,V}, T_{CZZ,U}, T_{A,U}, T_{BCS}$ and $T_{J, \text{est}}$. As reported in Table 4, the rejection rates of the proposed tests are 0.055 and 0.062, respectively, whereas those of $T_{CZZ,V}, T_{CZZ,U}, T_{A,U}, T_{BCS}$ and $T_{J, \text{est}}$ have different degrees of deviation from 0.05. Given the significant level $\alpha = 0.05$, our proposed tests provide the evidence that standardized genes from different groups are likely uncorrelated. This is confirmed by the Heat map in Figure 1. Similar phenomena are also found in Rothman et al. (2009), Cai and Liu (2011), and Xue et al. (2012).

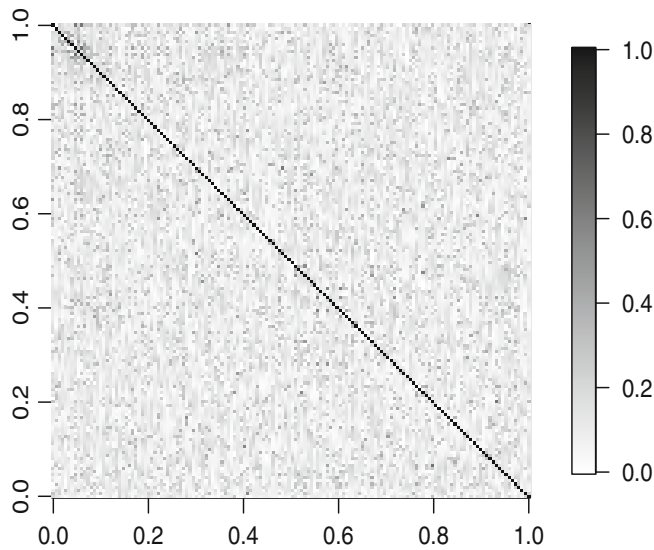


FIGURE 1 The heatmap of the absolute values of its sample correlation matrix.

6 | CONCLUSION

In this paper, we have studied some sphericity and identity U-tests for high-dimensional covariance matrices when the underlying population follows an elliptical distribution. We found that these location-invariant tests are not robust against elliptically high-dimensional data. We proposed modifications to make them robust with respect to elliptical distributions. We establish the universal (n, p) -asymptotic null distributions of the proposed test statistics, without requiring to specify either the kurtosis and skewness of data, or the growth rate between p and n . This enables our tests to be widely usable in practice. Numerical results show that our proposed tests can well control the type I error rates and achieve satisfactory power performances. Besides the problems we considered in this article, it is of interest to construct adjusted tests for bandedness and equality of high-dimensional covariance matrices under general multivariate procedures for nonnormal populations. These problems are related to the current work and yet left for our future investigations.

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REFERENCES

- Ahmad, M. R. (2016). On testing sphericity and identity of a covariance matrix with large dimensions. *Mathematical Methods of Statistics*, 25, 121–132.
- Anderson, T. W. (2003). *An introduction to multivariate statistical analysis* (3rd ed.). Wiley.
- Bai, Z., & Saranadasa, H. (1996). Effect of high dimension: By an example of a two sample problem. *Statistica Sinica*, 6, 311–329.
- Bai, Z. D., Jiang, D. D., Yao, J. F., & Zheng, S. R. (2009). Corrections to lrt on large dimensional covariance matrix by rmt. *The Annals of Statistics*, 37, 3822–3840.
- Bai, Z. D., & Silverstein, J. W. (2005). *Spectral analysis of large dimensional random matrices*. Scientific Press.
- Bickel, P. J., & Levina, E. (2008a). Regularized estimation of large covariance matrices. *The Annals of Statistics*, 36, 199–227.
- Bickel, P. J., & Levina, E. (2008b). Covariance regularization by thresholding. *The Annals of Statistics*, 36, 2577–2604.
- Birke, M., & Dette, H. (2005). A note on testing the covariance matrix for large dimension. *Statistics & Probability Letters*, 74, 281–289.
- Cai, T., & Liu, W. (2011). Adaptive thresholding for sparse covariance matrix estimation. *Journal of the American Statistical Association*, 106, 672–684.
- Cai, T. T., & Ma, Z. (2013). Optimal hypothesis testing for high dimensional covariance matrices. *Bernoulli*, 19, 2359–2388.
- Cai, T. T., Zhang, C. H., & Zhou, H. (2010). Optimal rates of convergence for covariance matrix estimation. *The Annals of Statistics*, 38, 2118–2144.
- Chen, S. X., & Qin, Y.-L. (2010). A two-sample test for high-dimensional data with applications to gene-set testing. *The Annals of Statistics*, 38, 808–835.
- Chen, S. X., Zhang, L. X., & Zhong, P. S. (2010). Tests for high-dimensional covariance matrices. *Journal of the American Statistical Association*, 105, 810–819.
- Cui, H., Guo, W., & Zhong, W. (2018). Test for high-dimensional regression coefficients using refitted cross-validation variance estimation. *Annals of Statistics*, 46, 958–988.
- Fang, K. T., Kotz, S., & Ng, K. W. (1990). *Symmetric multivariate and related distributions*. Chapman and Hall.
- Fisher, T. J. (2012). On testing for an identity covariance matrix when the dimensionality equals or exceeds the sample size. *Journal of Statistical Planning and Inference*, 142, 312–326.
- Fisher, T. J., Sun, X. Q., & Gallagher, C. M. (2010). A new test for sphericity of the covariance matrix for high dimensional data. *Journal of Multivariate Analysis*, 101, 2554–2570.
- Gao, J., Han, X., Pan, G., & Yang, Y. (2017). High dimensional correlation matrices: The central limit theorem and its applications. *Journal of the Royal Statistical Society: Series B*, 79, 677–693.
- Hall, P., & Heyde, C. C. (2014). *Martingale limit theory and its application*. Academic Press.
- Hallin, M., & Paindaveine, D. (2006). Semiparametrically efficient rank-based inference for shape i. optimal rank-based tests for sphericity. *The Annals of Statistics*, 34, 2707–2756.
- Himeno, T., & Yamada, T. (2014). Estimations for some functions of covariance matrix in high dimension under non-normality and its applications. *Journal of Multivariate Analysis*, 130, 27–44.
- Hu, J., Li, W., Liu, Z., & Zhou, W. (2019). High-dimensional covariance matrices in elliptical distributions with application to spherical test. *The Annals of Statistics*, 47, 527–555.
- Huang, J., Liu, N., Pourahmadi, M., & Liu, L. (2006). Covariance matrix selection and estimation via penalised normal likelihood. *Biometrika*, 93, 85–98.

- Ilmonen, P., & Paindaveine, D. (2011). Semiparametrically efficient inference based on signed ranks in symmetric independent component models. *The Annals of Statistics*, 39, 2448–2476.
- John, S. (1971). Some optimal multivariate tests. *Biometrika*, 58, 123–127.
- John, S. (1972). The distribution of a statistic used for testing sphericity of normal distributions. *Biometrika*, 59, 169–173.
- Khan, J., Wei, J. S., Ringner, M., Saal, L. H., Ladanyi, M., Westermann, F., Berthold, F., Schwab, M., Antonescu, C. R., Peterson, C., & Meltzer, P. S. (2001). Classification and diagnostic prediction of cancers using gene expression profiling and artificial neural networks. *Nature Medicine*, 7, 673–679.
- Ledoit, O., & Wolf, M. (2002). Some hypothesis tests for the covariance matrix when the dimension is large compared to the sample size. *The Annals of Statistics*, 30, 1081–1102.
- Li, W., & Yao, J. (2018). On structure testing for component covariance matrices of a high dimensional mixture. *Journal of the Royal Statistical Society: Series B*, 80, 293–318.
- Mardia, K. V., Jupp, P. E., & Mardia, K. (2000). *Directional statistics* (Vol. 2). Wiley Online Library.
- Möttönen, J., & Oja, H. (1995). Multivariate spatial sign and rank methods. *Journal of Nonparametric Statistics*, 5, 201–213.
- Muirhead, R. J., & Waternaux, C. M. (1980). Asymptotic distributions in canonical correlation analysis and other multivariate procedures for nonnormal populations. *Biometrika*, 67, 31–43.
- Nagao, H. (1973). On some test criteria for covariance matrix. *The Annals of Mathematical Statistics*, 1, 700–709.
- Paindaveine, D., & Verdebout, T. (2016). On high-dimensional sign tests. *Bernoulli*, 22, 1745–1769.
- Rothman, A. J., Levina, E., & Zhu, J. (2009). Generalized thresholding of large covariance matrices. *Journal of the American Statistical Association*, 104, 177–186.
- Rothman, A. J., Levina, E., & Zhu, J. (2010). A new approach to cholesky-based covariance regularization in high dimensions. *Biometrika*, 97, 539–550.
- Sirkiä, S., Taskinen, S., Oja, H., & Tyler, D. E. (2009). Tests and estimates of shape based on spatial signs and ranks. *Journal of Nonparametric Statistics*, 21, 155–176.
- Srivastava, M. (2005). Some tests concerning the covariance matrix in high-dimensional data. *Journal of the Japan Statistical Society*, 35, 251–272.
- Srivastava, M. S., Yanagihara, H., & Kubokawa, T. (2014). Tests for covariance matrices in high dimension with less sample size. *Journal of Multivariate Analysis*, 130, 289–309.
- Sugiura, N., & Nagao, H. (1968). Unbiasedness of some test criteria for the equality of one or two covariance matrices. *The Annals of Mathematical Statistics*, 39, 1686–1692.
- Tian, X. T., Lu, Y. T., & Li, W. M. (2015). A robust test for sphericity of high-dimensional covariance matrices. *Journal of Multivariate Analysis*, 141, 217–227.
- Wang, L., Peng, B., & Li, R. (2015). A high-dimensional nonparametric multivariate test for mean vector. *Journal of the American Statistical Association*, 110, 1658–1669.
- Wang, Q. W., & Yao, J. F. (2013). On the sphericity test with large-dimensional observations. *Electronic Journal of Statistics*, 7, 2164–2192.
- Xue, L., Ma, S., & Zou, H. (2012). Positive-definite l1-penalized estimation of large covariance matrices. *Journal of the American Statistical Association*, 107, 1480–1491.
- Yang, X., Zheng, X., & Chen, J. (2021). Testing high-dimensional covariance matrices under the elliptical distribution and beyond. *Journal of Econometrics*, 221, 409–423.
- Yin, Y., Zheng, S., & Zou, T. (2023). Central limit theorem of linear spectral statistics of high-dimensional sample correlation matrices. *Bernoulli*, 29, 984–1006.
- Zhang, R., Peng, L., & Wang, R. (2013). Tests for covariance matrix with fixed or divergent dimension. *The Annals of Statistics*, 41, 2075–2096.
- Zhong, P.-S., & Chen, S. X. (2011). Tests for high-dimensional regression coefficients with factorial designs. *Journal of the American Statistical Association*, 106, 260–274.
- Zou, C. L., Peng, L. H., Feng, L., & Wang, Z. J. (2014). Multivariate sign-based high-dimensional tests for sphericity. *Biometrika*, 101, 229–236.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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